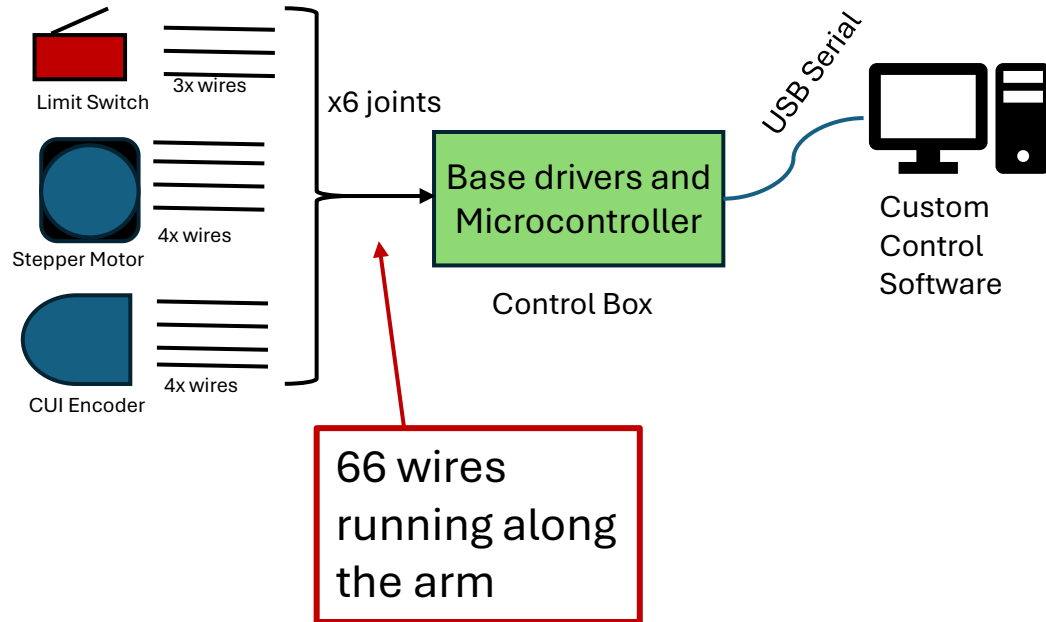


Retrofit stepper arm with CAN bus and Moveit

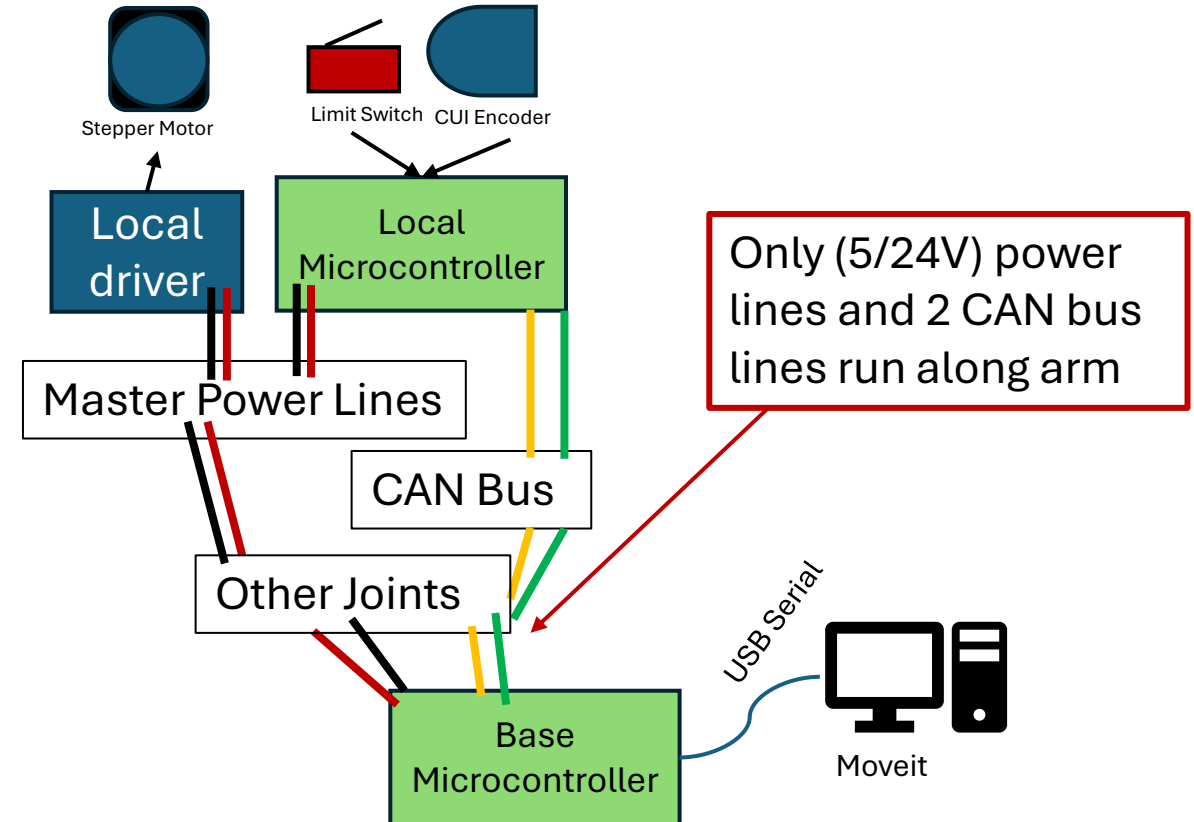
Notebook slides

Current Implementation



- Sensor read, serial interpretation and step commands all handled on the base controller

Proposed Implementation



- Joint level calibration and data handled locally on smaller microcontroller and driver
- Base controller only responsible for receiving incoming motion commands and delegating to joints over CAN bus

Current Implementation



Wires running along arm

Base drivers and Microcontroller

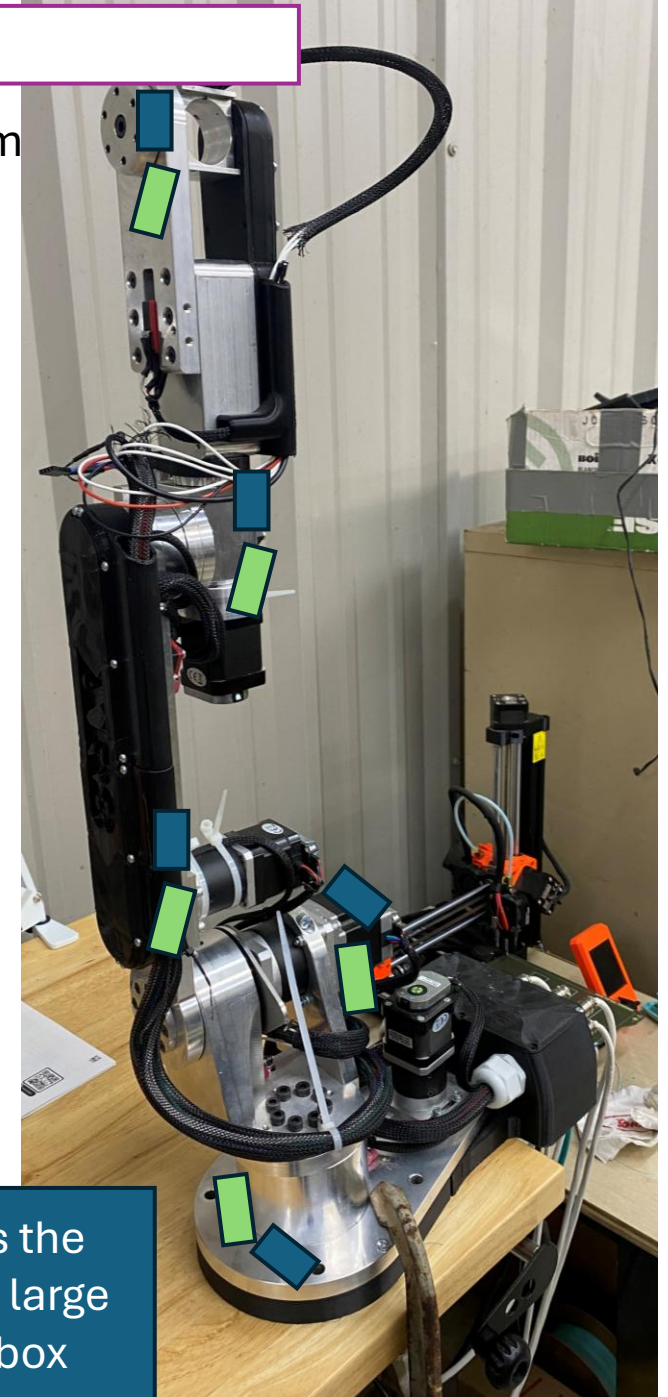
USB Serial

Proposed

Mounted Along arm

Local driver

Local Microcontroller



Removes the need for a large control box

Bill of materials

- Local Microcontrollers - Teensy 4.0
- Base Microcontroller – Teensy 4.1
- Stepper Drivers – (Testing 3 from Pololu)
- CAN transceivers

Development Steps

Hardware

1. Assemble single joint testbed
2. Test Pololu Drivers (A4988, DRV8825, DRV8434A)
3. Decide mounting options

Software

1. Read Encoder/Limit Switch
2. Step stepper w simple setpoint
3. Parse n' send CAN msgs
 1. Motor States? Pos, Vel, Accel motor profiles?
4. Base Microcontroller USB parse and CAN delegation
5. Moveit setup for kinematic trajecotries

Wiring Notes

Board Wiring diagram – driver to motor

Nema 14 Dual Shaft Stepper

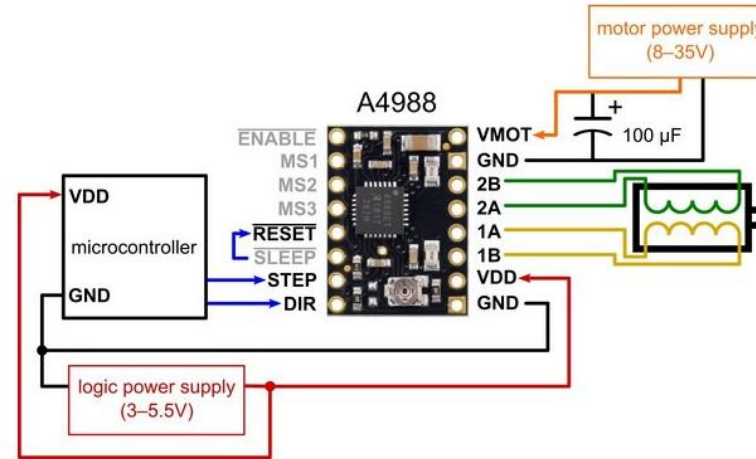
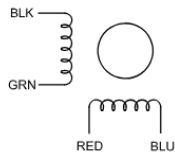
Rated Current/phase: 1.0A

Phase Resistance: 3.5 ohms

TYPE OF CONNECTION (EXTERN)		MOTOR	
PIN NO	BIPOLAR	LEADS	WINDING
1	A —	BLK	A
2	A\ —	GRN	A\
3	B —	RED	B
4	B\ —	BLU	B\

STEP 2 PHASE-Ex. ,
√ FACING MOUNTING END (X)

A	B	A\	B\	
+	+	-	-	
-	+	+	-	↓
-	-	+	+	↑
+	-	-	+	



Minimal wiring diagram for connecting a microcontroller to an A4988 stepper motor driver carrier (full-step mode).

Pololu [product](https://www.pololu.com/product/2128)

<https://www.pololu.com/product/2128>

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Wired Components

Driver	Motor
2B	A\ – GRN
2A	A – BLK
1A	B – RED
1B	B\ – BLU

Wiring diagram – Microcontroller Wiring

- TODOs
- Power Supplies (4 pins)
- Controller(2-4 pins)
- CAN transceiver (4pins)
- Encoder (4pins)

